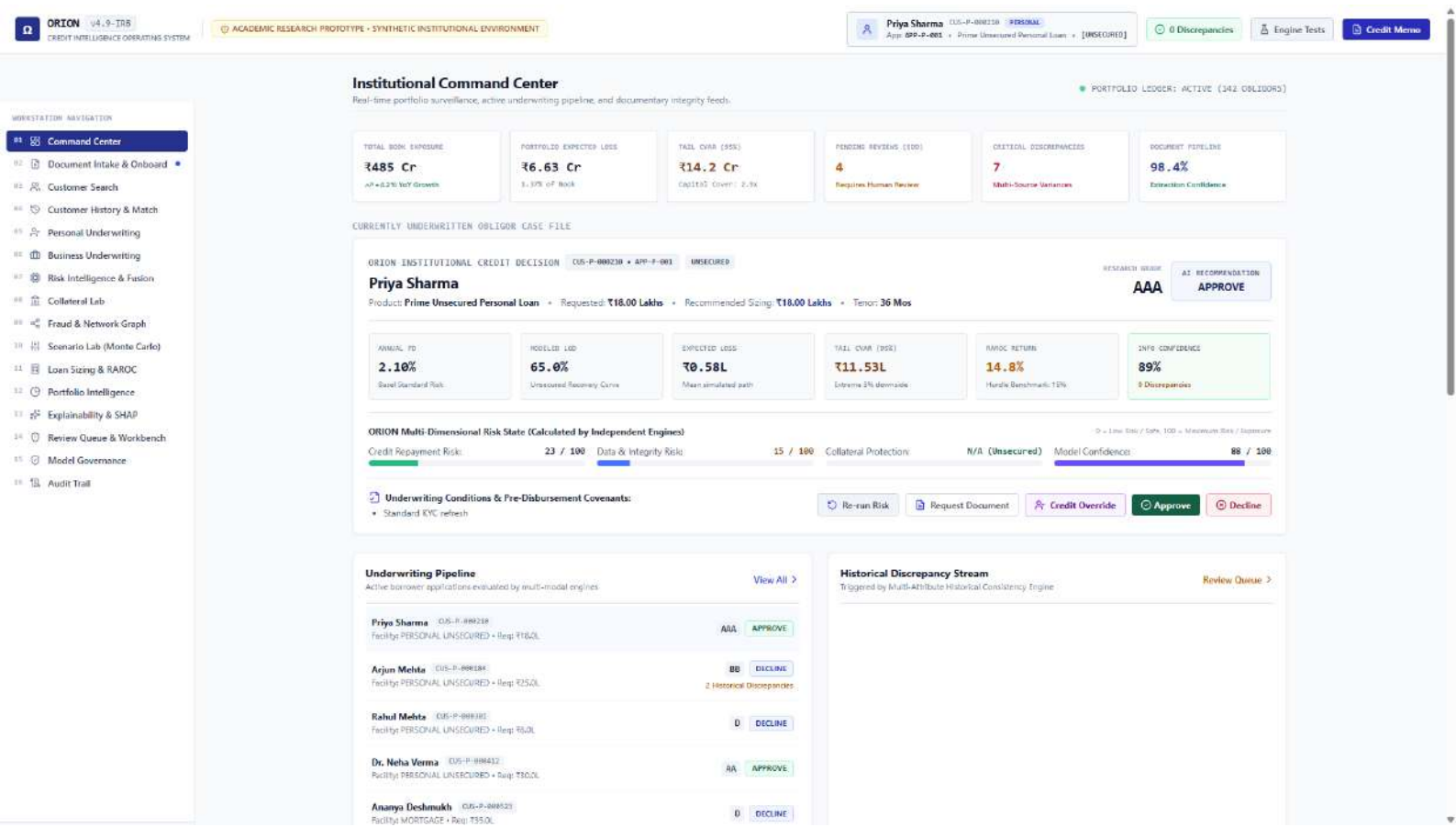
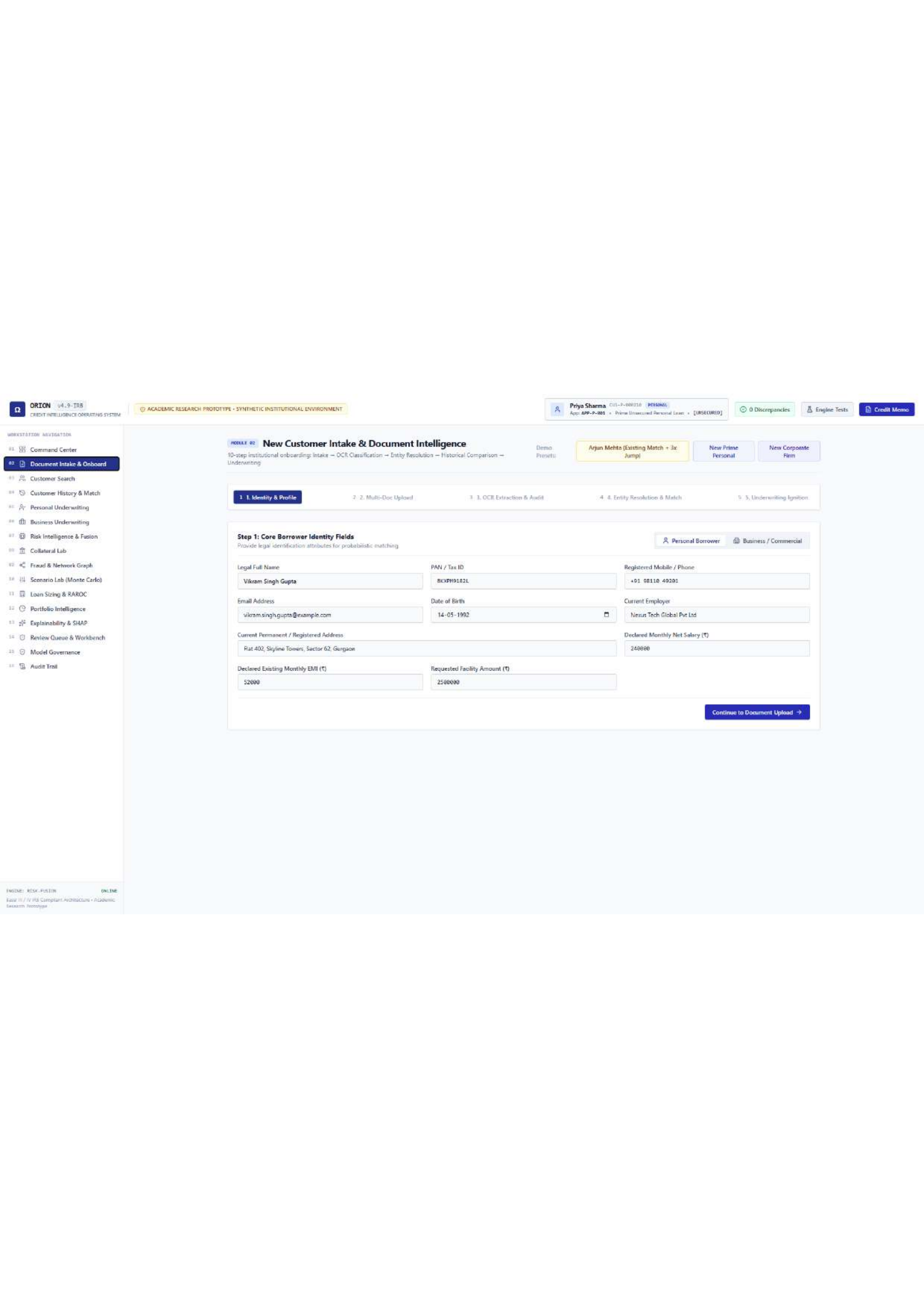
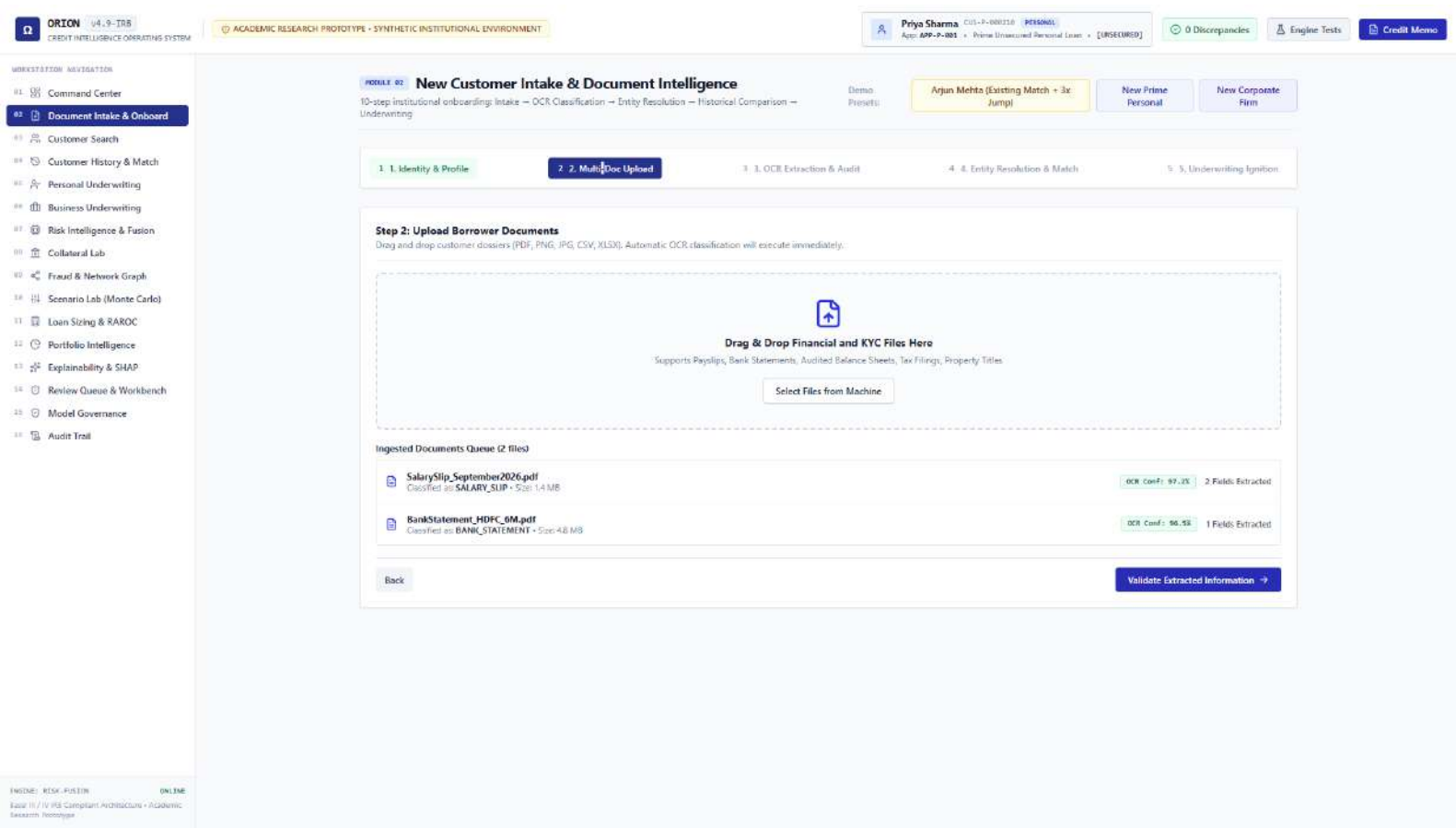
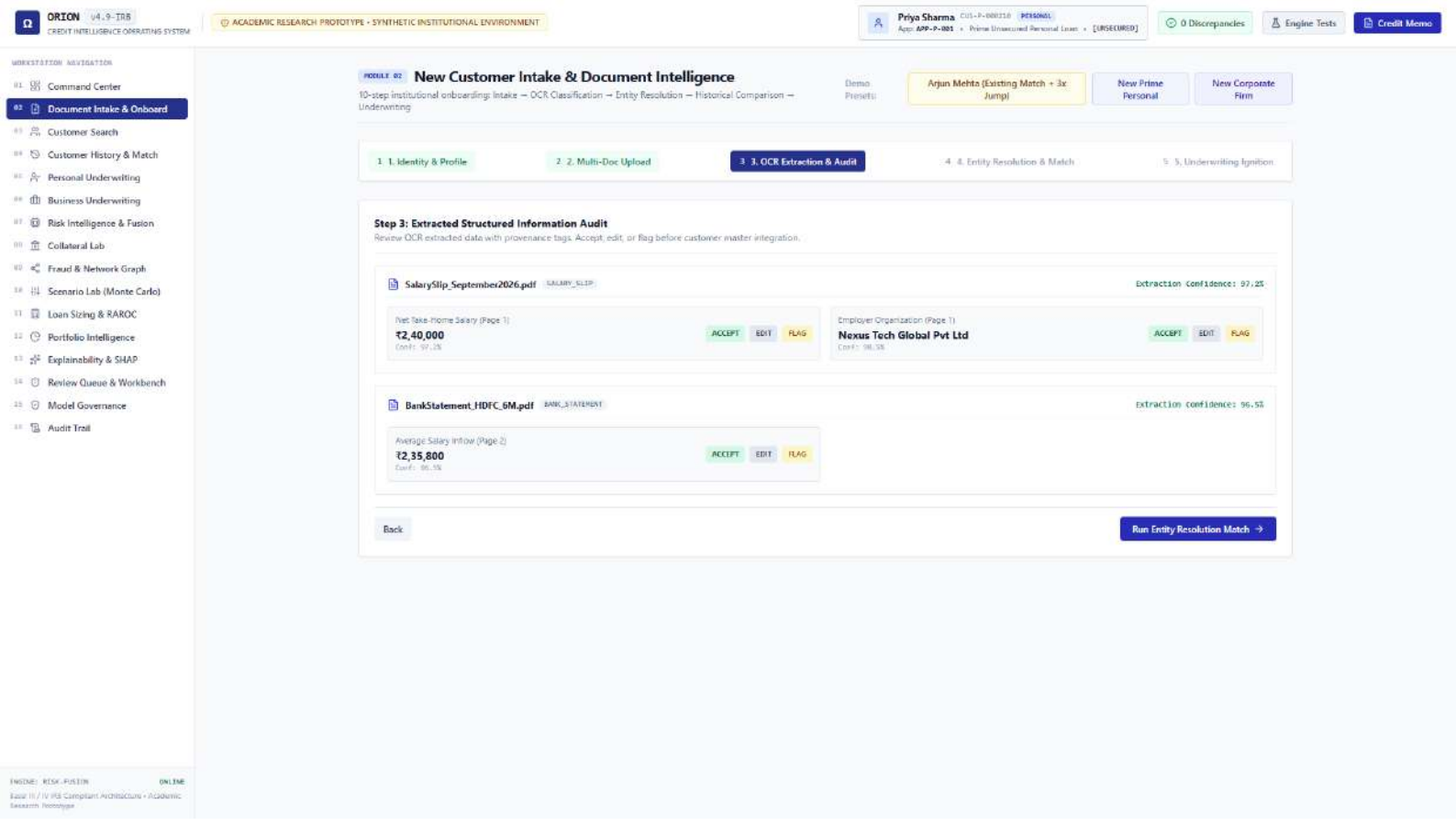
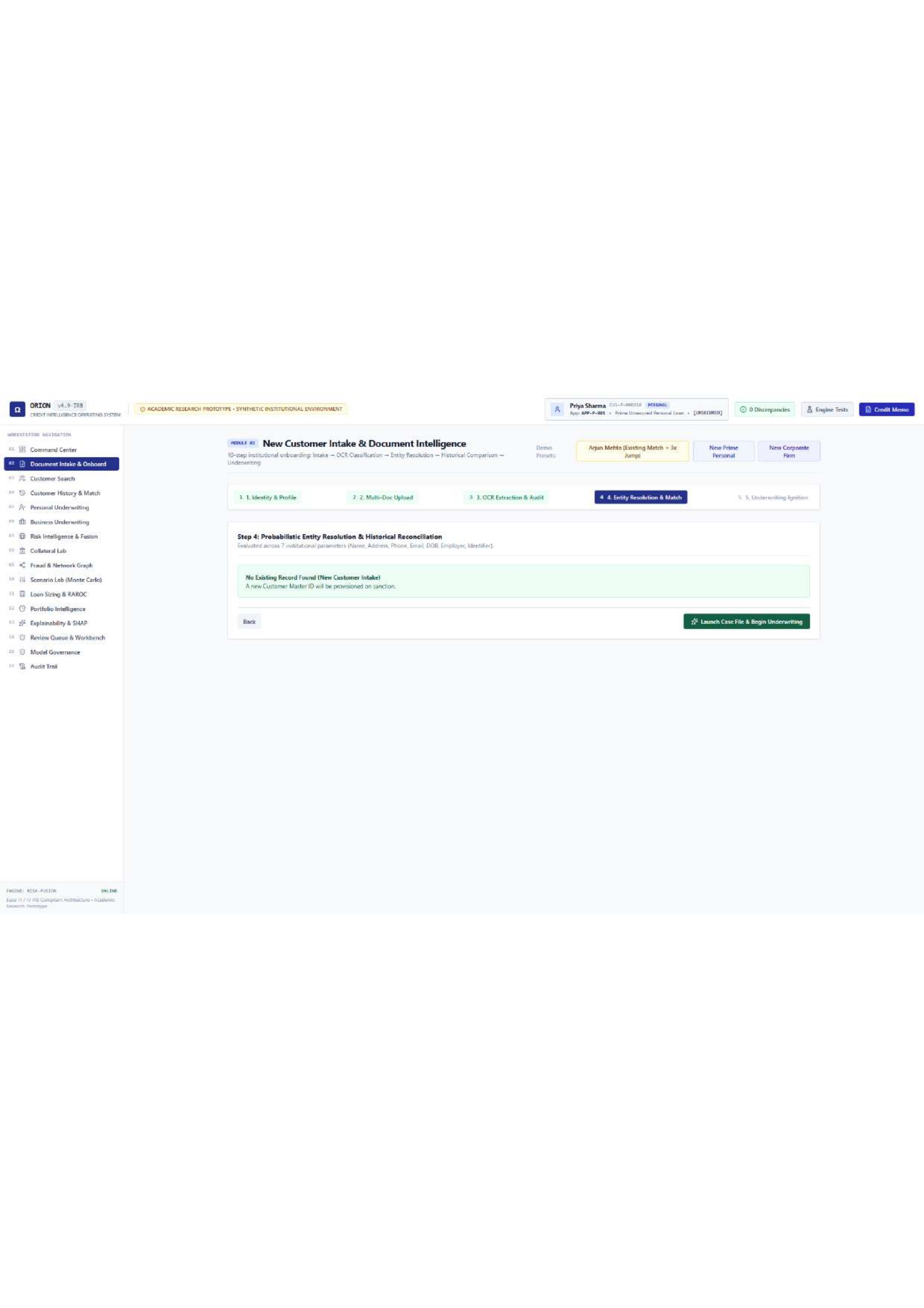
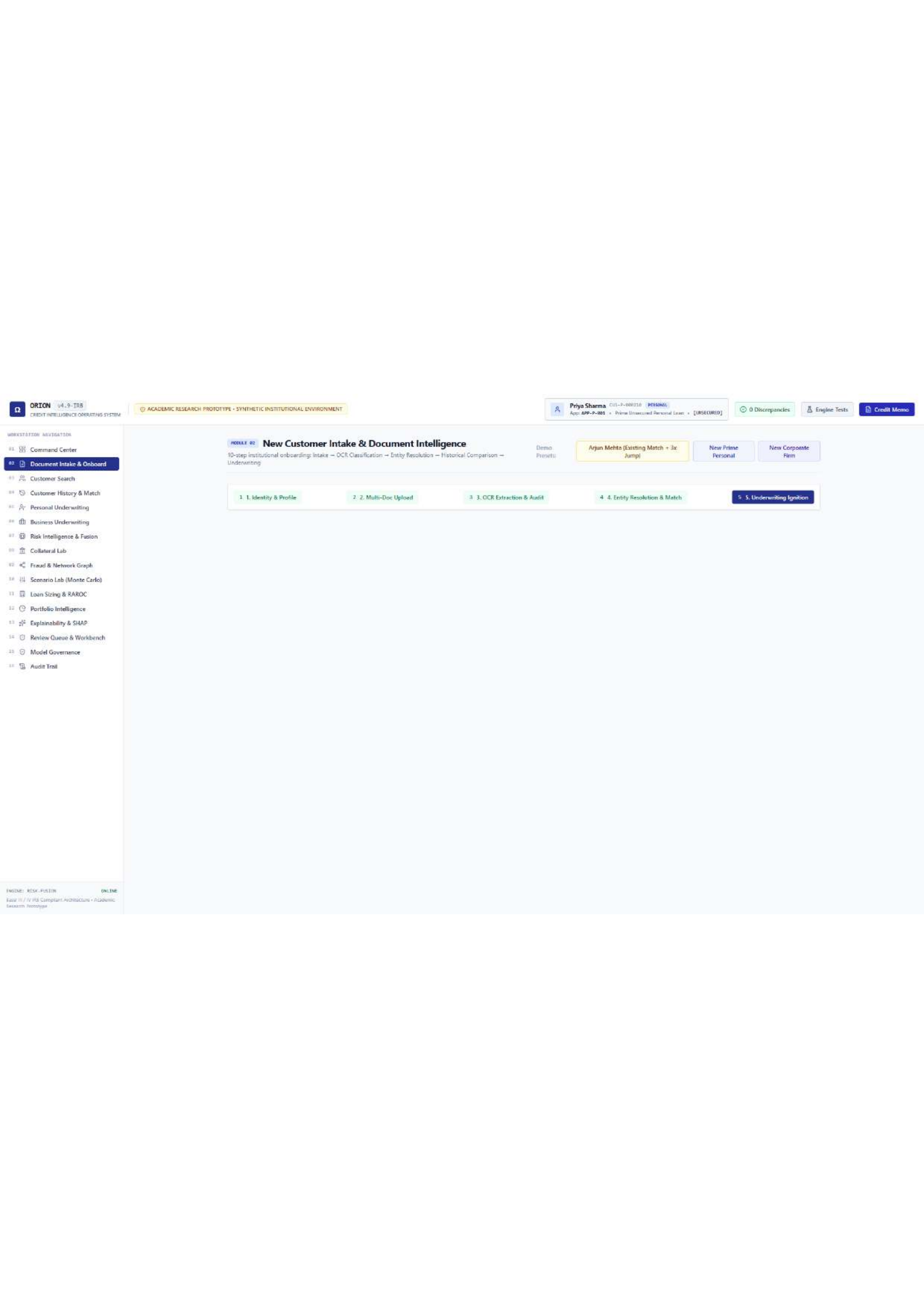

ORION v4.9-TRB
CREDIT INTELLIGENCE OPERATING SYSTEM
ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT
Priya Sharma CUS-P-000210 PERSONAL
App APP-P-001 · Prime Unsecured Personal Loan · [UNSECURED]
0 Discrepancies
Engine Tests
Credit Memo
Institutional Command Center
Real-time portfolio surveillance, active underwriting pipeline, and documentary integrity feeds.
PORTFOLIO LEDGER: ACTIVE (342 OBLIGORS)
WORKSTATION NAVIGATION
01 Command Center
02 Document Intake & Onboard
03 Customer Search
04 Customer History & Match
05 Personal Underwriting
06 Business Underwriting
07 Risk Intelligence & Fusion
08 Collateral Lab
09 Fraud & Network Graph
10 Scenario Lab (Monte Carlo)
11 Loan Sizing & RAROC
12 Portfolio Intelligence
13 Explainability & SHAP
14 Review Queue & Workbench
15 Model Governance
16 Audit Trail
TOTAL BOOK EXPOSURE
₹485 Cr
+4.2% YoY Growth
PORTFOLIO EXPECTED LOSS
₹6.63 Cr
1.37% of Book
TAIL CVAR (95%)
₹14.2 Cr
Capital Cover: 2.9x
PENDING REVIEWS (1DD)
4
Requires Human Review
CRITICAL DISCREPANCIES
7
Multi-Source Variances
DOCUMENT PIPELINE
98.4%
Extraction Confidence
CURRENTLY UNDERWRITTEN OBLIGOR CASE FILE
ORION INSTITUTIONAL CREDIT DECISION CUS-P-000210 · APP-P-001 UNSECURED
Priya Sharma
Product: Prime Unsecured Personal Loan · Requested: ₹18.00 Lakhs · Recommended Sizing: ₹18.00 Lakhs · Tenor: 36 Mos
RESEARCH GRADE
AAA
AI RECOMMENDATION
APPROVE
ANNUAL PD
2.10%
Basel Standard Risk
MODELED LGD
65.0%
Unsecured Recovery Curve
EXPECTED LOSS
₹0.58L
Mean simulated path
TAIL CVAR (95%)
₹11.53L
Extreme 5% downside
RAROC RETURN
14.8%
Hurdle Benchmark: 18%
INFO CONFIDENCE
89%
0 Discrepancies
ORION Multi-Dimensional Risk State (Calculated by Independent Engines)
0 = Low Risk / Safe, 100 = Maximum Risk / Exposure
Credit Repayment Risk: 23 / 100
Data & Integrity Risk: 15 / 100
Collateral Protection: N/A (Unsecured)
Model Confidence: 88 / 100
Underwriting Conditions & Pre-Disbursement Covenants:
Standard KYC refresh
Re-run Risk
Request Document
Credit Override
Approve
Decline
Underwriting Pipeline
Active borrower applications evaluated by multi-modal engines
View All
Priya Sharma CUS-P-000210
Facility: PERSONAL UNSECURED · Req: ₹18.0L
AAA APPROVE
Arjun Mehta CUS-P-000184
Facility: PERSONAL UNSECURED · Req: ₹25.0L
BB DECLINE
2 Historical Discrepancies
Rahul Mehta CUS-P-000101
Facility: PERSONAL UNSECURED · Req: ₹8.0L
D DECLINE
Dr. Neha Verma CUS-P-000412
Facility: PERSONAL UNSECURED · Req: ₹30.0L
AA APPROVE
Ananya Deshmukh CUS-P-000523
Facility: MORTGAGE · Req: ₹95.0L
D DECLINE
Historical Discrepancy Stream
Triggered by Multi-Attribute Historical Consistency Engine
Review Queue

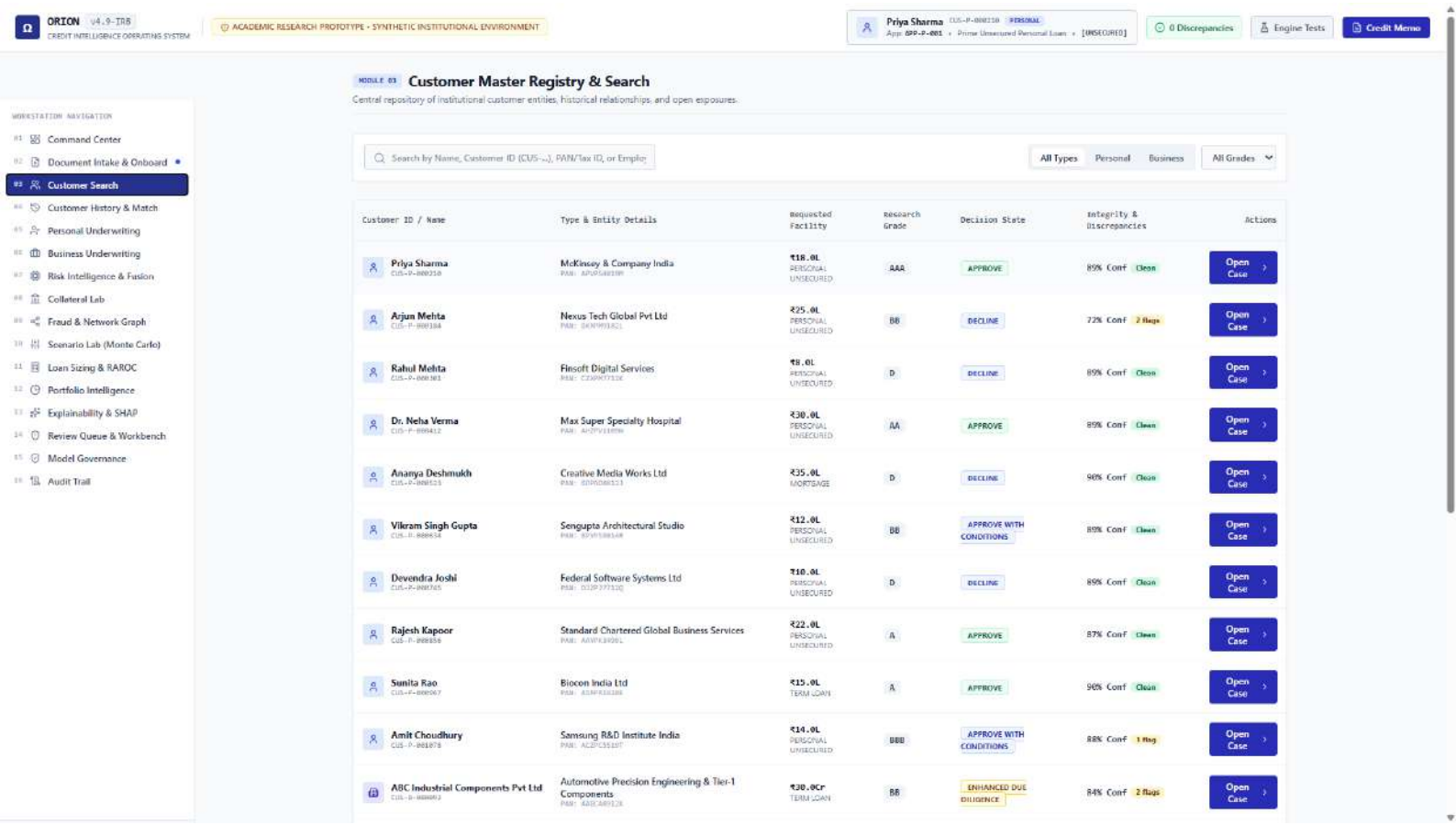

ORION v4.9-TRB
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ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT
Priya Sharma CUS-P-000210 PERSONAL
App: APP-P-001 · Prime Unsecured Personal Loan · [UNSECURED]
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Credit Memo
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16 Audit Trail
ENGINE: RISK-FUSION ONLINE
Basel III / IV IRB Compliant Architecture · Academic Research Prototype
MODULE 02
New Customer Intake & Document Intelligence
10-step institutional onboarding: Intake → OCR Classification → Entity Resolution → Historical Comparison → Underwriting
Demo Presets:
Arjun Mehta (Existing Match + 3x Jump)
New Prime Personal
New Corporate Firm
1. Identity & Profile
2. Multi-Doc Upload
3. OCR Extraction & Audit
4. Entity Resolution & Match
5. Underwriting Ignition
Step 1: Core Borrower Identity Fields
Provide legal identification attributes for probabilistic matching.
Personal Borrower
Business / Commercial
Legal Full Name
Vikram Singh Gupta
PAN / Tax ID
BKXPH9182L
Registered Mobile / Phone
+91 98110 49201
Email Address
vikram.singh.gupta@example.com
Date of Birth
14-05-1992
Current Employer
Nexus Tech Global Pvt Ltd
Current Permanent / Registered Address
Flat 402, Skyline Towers, Sector 62, Gurgaon
Declared Monthly Net Salary (₹)
240000
Declared Existing Monthly EMI (₹)
52000
Requested Facility Amount (₹)
2500000
Continue to Document Upload →


ORION v4.9-TRB
CREDIT INTELLIGENCE OPERATING SYSTEM
ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT
Priya Sharma
0 Discrepancies
Engine Tests
Credit Memo
WORKSTATION NAVIGATION
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Step 2: Upload Borrower Documents
Drag and drop customer dossiers (PDF, PNG, JPG, CSV, XLSX). Automatic OCR classification will execute immediately.
Drag & Drop Financial and KYC Files Here
Supports Payslips, Bank Statements, Audited Balance Sheets, Tax Filings, Property Titles
Select Files from Machine
Ingested Documents Queue (2 files)
SalarySlip_September2026.pdf
Classified as: SALARY_SLIP • Size: 1.4 MB
OCR Conf: 97.2%
2 Fields Extracted
BankStatement_HDFC_6M.pdf
Classified as: BANK_STATEMENT • Size: 4.8 MB
OCR Conf: 96.5%
1 Fields Extracted
Back
Validate Extracted Information →


ORION v4.9-TRB
CREDIT INTELLIGENCE OPERATING SYSTEM
ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT
Priya Sharma CUS-P-000210 PERSONAL
App: APP-P-001 · Prime Unsecured Personal Loan · [UNSECURED]
0 Discrepancies
Engine Tests
Credit Memo
WORKSTATION NAVIGATION
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Arjun Mehta (Existing Match + 3x Jump)
New Prime Personal
New Corporate Firm
1. Identity & Profile
2. Multi-Doc Upload
3. OCR Extraction & Audit
4. Entity Resolution & Match
5. Underwriting Ignition
Step 3: Extracted Structured Information Audit
Review OCR extracted data with provenance tags. Accept, edit, or flag before customer master integration.
SalarySlip_September2026.pdf
SALARY_SLIP
Extraction Confidence: 97.2%
Net Take-Home Salary (Page 1)
₹2,40,000
Conf: 97.2%
ACCEPT
EDIT
FLAG
Employer Organization (Page 1)
Nexus Tech Global Pvt Ltd
Conf: 98.5%
ACCEPT
EDIT
FLAG
BankStatement_HDFC_6M.pdf
BANK_STATEMENT
Extraction Confidence: 96.5%
Average Salary Inflow (Page 2)
₹2,35,800
Conf: 96.5%
ACCEPT
EDIT
FLAG
Back
Run Entity Resolution Match →


ORION v4.9-TRB
CREDIT INTELLIGENCE OPERATING SYSTEM
ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT
Priya Sharma CUS-P-000210 PERSONAL
App: APP-P-001 · Prime Unsecured Personal Loan · [UNSECURED]
0 Discrepancies
Engine Tests
Credit Memo
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Basel III / IV IRB Compliant Architecture · Academic Research Prototype
MODULE 02
New Customer Intake & Document Intelligence
10-step institutional onboarding: Intake → OCR Classification → Entity Resolution → Historical Comparison → Underwriting
Demo Presets:
Arjun Mehta (Existing Match + 3x Jump)
New Prime Personal
New Corporate Firm
1. Identity & Profile
2. Multi-Doc Upload
3. OCR Extraction & Audit
4. Entity Resolution & Match
5. Underwriting Ignition
Step 4: Probabilistic Entity Resolution & Historical Reconciliation
Evaluated across 7 institutional parameters (Name, Address, Phone, Email, DOB, Employer, Identifier).
No Existing Record Found (New Customer Intake)
A new Customer Master ID will be provisioned on sanction.
Back
Launch Case File & Begin Underwriting


ORION v4.9-TRB
CREDIT INTELLIGENCE OPERATING SYSTEM
ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT
Priya Sharma CUS-P-000210 PERSONAL
App: APP-P-001 · Prime Unsecured Personal Loan · [UNSECURED]
0 Discrepancies
Engine Tests
Credit Memo
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ENGINE: RISK-FUSION ONLINE
Basel III / IV IRB Compliant Architecture · Academic Research Prototype
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10-step institutional onboarding: Intake → OCR Classification → Entity Resolution → Historical Comparison → Underwriting
Demo Presets:
Arjun Mehta (Existing Match + 3x Jump)
New Prime Personal
New Corporate Firm
1 1. Identity & Profile
2 2. Multi-Doc Upload
3 3. OCR Extraction & Audit
4 4. Entity Resolution & Match
5 5. Underwriting Ignition


ORION v4.9-TRB
CREDIT INTELLIGENCE OPERATING SYSTEM
ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT
Priya Sharma CUS-P-000350 PERSONAL
App SPP-P-001 · Prime Unsecured Personal Loan · [UNSECURED]
0 Discrepancies
Engine Tests
Credit Memo
MODULE 03
Customer Master Registry & Search
Central repository of institutional customer entities, historical relationships, and open exposures.
WORKSTATION NAVIGATION
01 Command Center
02 Document Intake & Onboard
03 Customer Search
04 Customer History & Match
05 Personal Underwriting
06 Business Underwriting
07 Risk Intelligence & Fusion
08 Collateral Lab
09 Fraud & Network Graph
10 Scenario Lab (Monte Carlo)
11 Loan Sizing & RAROC
12 Portfolio Intelligence
13 Explainability & SHAP
14 Review Queue & Workbench
15 Model Governance
16 Audit Trail
Search by Name, Customer ID (CUS-...), PAN/Tax ID, or Employ
All Types
Personal
Business
All Grades
Customer ID / Name
Type & Entity Details
Requested Facility
Research Grade
Decision State
Integrity & Discrepancies
Actions
Priya Sharma
CUS-P-000350
McKinsey & Company India
₹18.0L
PERSONAL UNSECURED
AAA
APPROVE
89% Conf
Clean
Open Case
Arjun Mehta
CUS-P-000384
Nexus Tech Global Pvt Ltd
₹25.0L
PERSONAL UNSECURED
BB
DECLINE
72% Conf
2 flags
Open Case
Rahul Mehta
CUS-P-000383
Finsoft Digital Services
₹8.0L
PERSONAL UNSECURED
D
DECLINE
89% Conf
Clean
Open Case
Dr. Neha Verma
CUS-P-000412
Max Super Specialty Hospital
₹30.0L
PERSONAL UNSECURED
AA
APPROVE
89% Conf
Clean
Open Case
Ananya Deshmukh
CUS-P-000523
Creative Media Works Ltd
₹35.0L
MORTGAGE
D
DECLINE
96% Conf
Clean
Open Case
Vikram Singh Gupta
CUS-P-000438
Sengupta Architectural Studio
₹12.0L
PERSONAL UNSECURED
BB
APPROVE WITH CONDITIONS
89% Conf
Clean
Open Case
Devendra Joshi
CUS-P-000765
Federal Software Systems Ltd
₹10.0L
PERSONAL UNSECURED
D
DECLINE
89% Conf
Clean
Open Case
Rajesh Kapoor
CUS-P-000858
Standard Chartered Global Business Services
₹22.0L
PERSONAL UNSECURED
A
APPROVE
87% Conf
Clean
Open Case
Sunita Rao
CUS-P-000967
Biocon India Ltd
₹15.0L
TERM LOAN
A
APPROVE
96% Conf
Clean
Open Case
Amit Choudhury
CUS-P-001078
Samsung R&D Institute India
₹14.0L
PERSONAL UNSECURED
BBB
APPROVE WITH CONDITIONS
88% Conf
1 flag
Open Case
ABC Industrial Components Pvt Ltd
Automotive Precision Engineering & Tier-1 Components
₹30.0Cr
TERM LOAN
BB
ENHANCED DUE DILIGENCE
84% Conf
2 flags
Open Case

WORKSTATION NAVIGATION
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ENGINE: RISK-FUSION ONLINE

MODULE 04

Customer Master History & Record Reconciliation

Immutable longitudinal profile history, past loan performance ledgers, and current-versus-historical comparator.

Customer ID: CUS-P-000210

Side-by-Side Profile Comparator

Direct verification comparing current intake against institutional historical baseline

No Pending Variances

Profile Attribute	Current Application (2026)	Historical Profile Baseline	Variance Velocity	Audit Source Evidence
Monthly Net Income	₹4,20,000	₹3,80,000		
Permanent / Operating Address	Tower 4, Apt 1802, Prestige Golfshire, Bengaluru, Karnataka 560001	Prestige Golfshire, Bengaluru	Address Changed	Current Intake Form vs APP-P-000
Current Employer	McKinsey & Company India	McKinsey & Company India		
Existing Monthly Debt (EMI)	₹45,000	₹45,000		
Declared Collateral Security	Clean / Unsecured Facility	Unsecured Personal Facility		

Longitudinal Customer Master Timeline

Audited sequence of customer onboarding events, facilities sanctioned, and profile snapshots

SEP 2026 CURRENT APPLICATION
New Facility Requested: ₹10.0 Lakhs (Prime Unsecured Personal Loan)
Declared Monthly Take-Home: ₹4,20,000 • Current Address: Tower 4, Apt 1802, Prestige Golfshire, Bengaluru, Karnataka 560001

APR 2024 HISTORICAL AUDIT SNAPSHOT Grade AAA
Reported Monthly Income: ₹3,80,000/month • Employer: McKinsey & Company India • Permanent Address: Prestige Golfshire, Bengaluru

APR 2024 LOAN DISBURSEMENT
Sanctioned Personal Loan ₹15.0 Lakhs (LN-PS-2024) • Repayment status: ACTIVE, 0 DPD across 36 months

OCTOBER 2023 CUSTOMER ONBOARDED
Initial KYC verification completed; provisioned Customer Master ID CUS-P-000210

Historical Institutional Credit Facilities

Historical lending relationships retrieved from internal institutional core-banking ledger

Loan ID	Product Type	Sanction Date	Amount	Tenure / Rate	Outstanding Balance	Max DPD	Status	Human Override
LN-PS-2024	Personal Loan	2024-04-15	₹15.0L	36M @ 9.5%	₹4.2L	0 DPD	ACTIVE	None

ORION v4.9-TRB
CREDIT INTELLIGENCE OPERATING SYSTEM

ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT

Priya Sharma CUS-P-000210 PERSONAL
App APP-P-001 · Prime Unsecured Personal Loan · [UNSECURED]

0 Discrepancies | Engine Tests | Credit Memo

WORKSTATION NAVIGATION

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MODULE 05

Personal Credit Underwriting & ML Risk Intelligence

Credit bureau tradelines, Fixed Obligation to Income Ratio (FOIR), transaction volatility, and real-time ML inference.

Borrower: Priya Sharma (CUS-P-000210)

Live Supervised ML Pipeline Execution & Calibration Trace

STATUS: REAL-TIME INFERENCE ACTIVE Model: ORION-XGBoost-IRB v4.8 (Dec)

Baseline Logistic PD	Random Forest PD	Tuned XGBoost PD	Calibrated PD & Grade
0.78%	1.00%	0.80%	0.78% (AAA)
Linear logit benchmark	150 bagging trees	Production candidate	Platt Signoid (ECE: 0.048)

END-TO-END TRACEABLE PIPELINE EXECUTION PATH:

Raw Data (18 fields) → Validation (Score: 98) → Feature Eng. (14 Predictors) → Preprocessing (Robust Σ) → XGBoost Inference → Platt Calibration → Decision: APPROVE

Product Collateral Structure: Prime Unsecured Personal Loan

CLEAN UNSECURED FACILITY

Regulatory Retail Policy: Pledged collateral valuation is recorded as ₹0 (UNSECURED). This facility is underwritten strictly against verified monthly salary cash inflows and bureau debt service capacity. *No missing-collateral warning is generated.*

Interactive Income Shock Stress Testing Knob

Shock Applied: -0% Monthly Take-Home

-0% | -15% | -25% | -40%

Effective Income:	Tested FOIR:	Stressed PD:	Stressed Capacity:
₹4,20,000	25%	2.10%	₹49.3L

Comprehensive Credit Bureau Dossier

Bureau Score: 812 (PRIME)

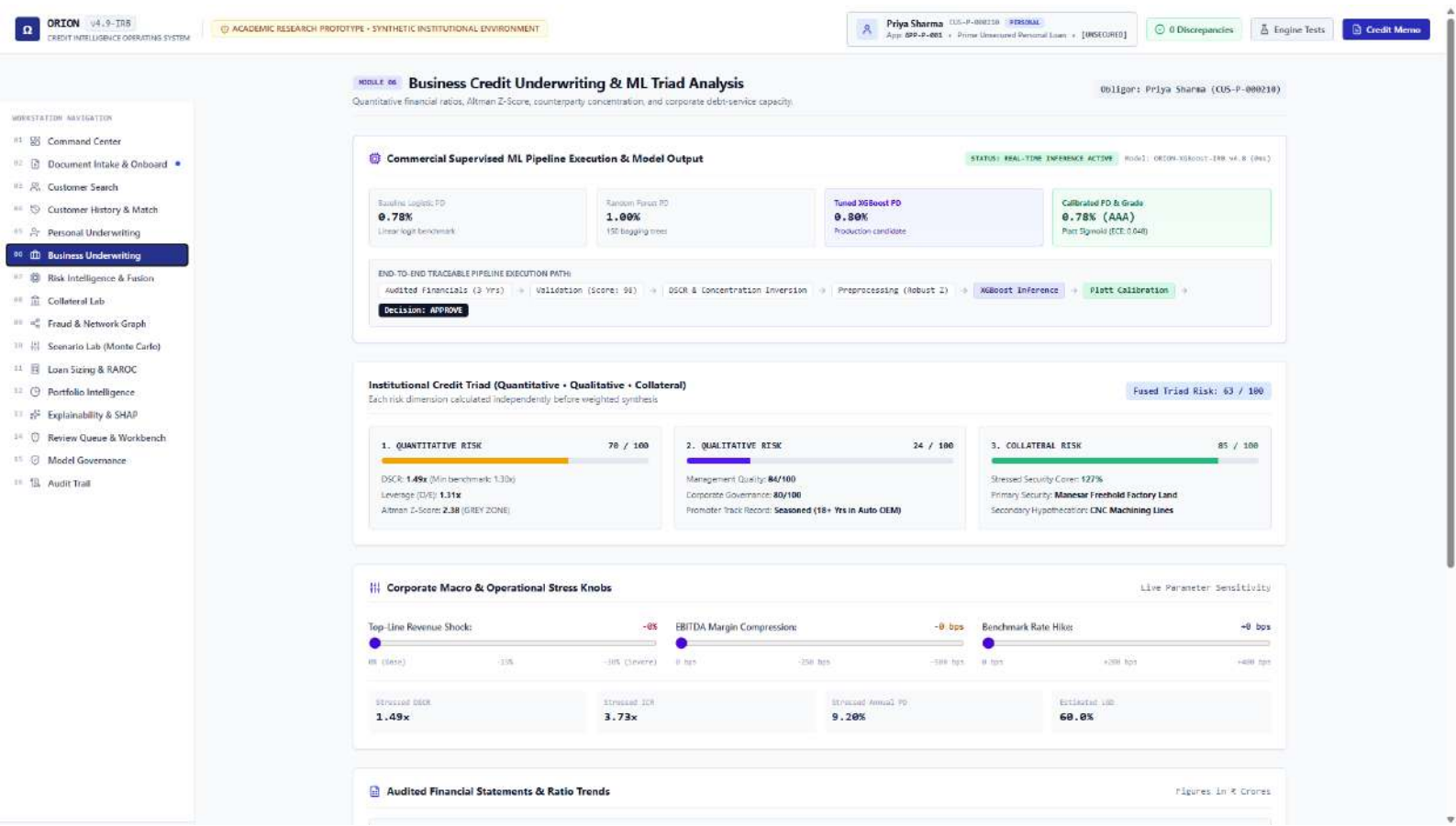
REVOLVING UTILIZATION	CREDIT AGE & TENURE	RECENT INQUIRIES (6M)	DELINQUENCY (DPD 30+)
18%	9.4 Years	0 Inquiries	0 Accounts
Risk: LOW	Seasoned Repayment	No Enquiries	Flawless Track Record

Active Credit Bureau Tradelines

Lender Institution	Account Type	Sanction Date	Sanction Limit	Current Balance	Monthly EMI	Conduct
ORION Bank	PERSONAL LOAN	2024-04-15	₹15.0L	₹4.2L	₹48,000	0 DPD (Current)

Banking Cash Flow & Transaction Stability

Stability Index: 80 / 100


ORION v4.9-TRB
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ACADEMIC RESEARCH PROTOTYPE · SYNTHETIC INSTITUTIONAL ENVIRONMENT
Priya Sharma CUS-P-000210 PERSONAL
App APP-P-001 · Prime Unsecured Personal Loan · [UNSCORED]
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MODULE 06
Business Credit Underwriting & ML Triad Analysis
Quantitative financial ratios, Altman Z-Score, counterparty concentration, and corporate debt-service capacity.
Obligor: Priya Sharma (CUS-P-000210)
Commercial Supervised ML Pipeline Execution & Model Output
STATUS: REAL-TIME INFERENCE ACTIVE
Model: ORION-XGBoost-IRB v4.8 (Dec)
Baseline Logistic PD
0.78%
Linear logit benchmark
Random Forest PD
1.00%
150 bagging trees
Tuned XGBoost PD
0.80%
Production candidate
Calibrated PD & Grade
0.78% (AAA)
Platt Sigmoid (ECE: 0.048)
END-TO-END TRACEABLE PIPELINE EXECUTION PATH:
Audited Financials (3 Yrs)
Validation (Score: 98)
DSCR & Concentration Inversion
Preprocessing (Robust Σ)
XGBoost Inference
Platt Calibration
Decision: APPROVE
Institutional Credit Triad (Quantitative + Qualitative + Collateral)
Each risk dimension calculated independently before weighted synthesis
Fused Triad Risk: 63 / 100
1. QUANTITATIVE RISK
70 / 100
DSCR: 1.49x (Min benchmark: 1.30x)
Leverage (D/E): 1.31x
Altman Z-Score: 2.38 (GREY ZONE)
2. QUALITATIVE RISK
24 / 100
Management Quality: 84/100
Corporate Governance: 80/100
Promoter Track Record: Seasoned (18+ Yrs in Auto OEM)
3. COLLATERAL RISK
85 / 100
Stressed Security Cover: 127%
Primary Security: Manesar Freehold Factory Land
Secondary Hypothecation: CNC Machining Lines
Corporate Macro & Operational Stress Knobs
Live Parameter Sensitivity
Top-Line Revenue Shock:
-0%
0% (Base)
-15%
-30% (Severe)
EBITDA Margin Compression:
-0 bps
0 bps
-250 bps
-500 bps
Benchmark Rate Hike:
+0 bps
0 bps
+200 bps
+400 bps
Stressed DSCR
1.49x
Stressed ICR
3.73x
Stressed Annual PD
9.20%
Estimated LGD
60.0%
Audited Financial Statements & Ratio Trends
Figures in ₹ Crores

WORKSTATION NAVIGATION
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MODULE 07

Risk Intelligence & Multi-Modal Fusion Engine

Grounded synthesis separating fundamental Credit Default Risk from Document / Data Integrity Risk.

Grade: AAA • Model Conf: 88%

Separation of Fundamental Credit Risk vs Data Integrity Risk

Preventing documentation variances from being conflated with intrinsic borrower bankruptcy

Orion Decoupled Risk Framework

DIMENSION 1: CREDIT RISK (CAPACITY & WILL TO REPAY) 23 / 100

- Annual Probability of Default (PD): **2.10%**
- Loss Given Default (LGD): **65.0%**
- Debt-to-Income / DSCR Headroom: **43% FOIR**
- Assessment: **Financially solvent with viable debt-servicing buffer**

DIMENSION 2: DATA / INTEGRITY RISK (INFORMATION CONFIDENCE) 15 / 100

- Overall Information Confidence: **89%**
- Pending Discrepancy Items: **0 item(s)**
- Primary Drivers: **Consistent documentary baseline — No variance flags**
- Action Mandate: **Proceed with standard credit risk adjudication**

ORION Information Confidence Score Breakdown

Granular trust metrics across identity, income, employment, address, debt obligations, and collateral

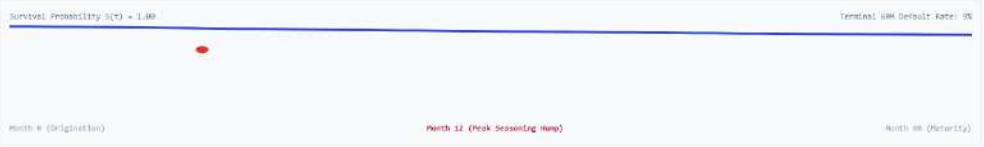
Composite Index: 89%

IDENTITY CONF.	INCOME CONF.	EMPLOYMENT CONF.	ADDRESS CONF.	DEBT CONF.	COLLATERAL CONF.
96%	88%	90%	85%	88%	100%
Biometric / PAN	Salary / Turnover	HR / Industry	KYC / Bureau Multiplicity	Bureau Tradelines	Valuation Deed

Survival Analysis & Time-to-Default Term Structure (1 - 60 Months)

Nelson-Aalen / Cox proportional hazards model capturing loan seasoning dynamics

Median Time to Default: 60 Months



"Hazard rate peaks at Month 12 due to credit seasoning dynamics. Projected 5-year cumulative default probability is 9%."

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MODULE 08

Collateral Lab & Valuation Intelligence

Asset haircuts, distressed liquidation values, second-charge encumbrance checks, and duplicate collateral alerts.

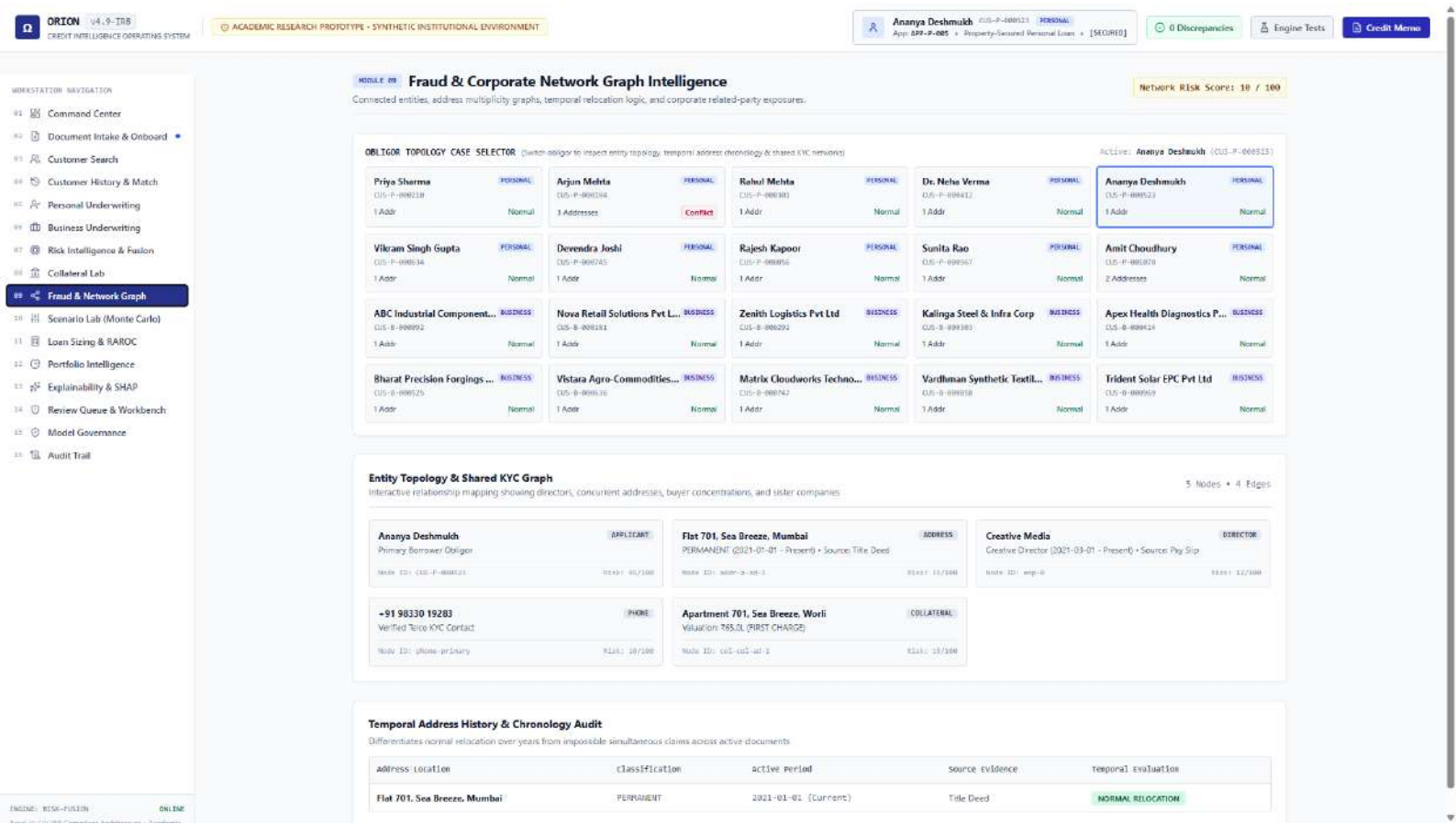
Stressed Security Cover: 149%

DOCUMENTED FAIR MARKET VALUE	STRESSED REALIZABLE VALUE	FORCED LIQUIDATION VALUE	STRESSED LTV RATIO
₹0.65 Cr	₹0.52 Cr	₹0.46 Cr	67.3%
Approved Valuer Report	After standard institutional haircut	90-day distress auction floor	Well-Secured Facility

Pledged Asset Schedule & Enforceability Audit

Granular breakdown of immovable properties, industrial plant machinery, and registered charge status

Asset Description	Category	Documented FMV	Haircut %	Stressed Value	Charge / Lien Priority	History Check	Verification
Apartment 701, Sea Breeze, Worli	RESIDENTIAL REAL ESTATE	₹65.0L	-20%	₹52.0L	FIRST CHARGE	ORIGINAL PLEDGE	Verified


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App: APP-P-005 · Property-Secured Personal Loan · [SECURED]
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ENGINE: RISK-FUSION ONLINE
MODULE 09
Fraud & Corporate Network Graph Intelligence
Connected entities, address multiplicity graphs, temporal relocation logic, and corporate related-party exposures.
Network Risk Score: 10 / 100
OBLIGOR TOPOLOGY CASE SELECTOR (Switch obligor to inspect entity topology, temporal address chronology & shared KYC networks)
Active: Ananya Deshmukh (CUS-P-000513)
Priya Sharma PERSONAL CUS-P-000210 1 Addr Normal
Arjun Mehta PERSONAL CUS-P-000384 3 Addresses Conflict
Rahul Mehta PERSONAL CUS-P-000301 1 Addr Normal
Dr. Neha Verma PERSONAL CUS-P-000412 1 Addr Normal
Ananya Deshmukh PERSONAL CUS-P-000523 1 Addr Normal
Vikram Singh Gupta PERSONAL CUS-P-000634 1 Addr Normal
Devendra Joshi PERSONAL CUS-P-000745 1 Addr Normal
Rajesh Kapoor PERSONAL CUS-P-000856 1 Addr Normal
Sunita Rao PERSONAL CUS-P-000967 1 Addr Normal
Amit Choudhury PERSONAL CUS-P-001078 2 Addresses Normal
ABC Industrial Component... BUSINESS CUS-B-000092 1 Addr Normal
Nova Retail Solutions Pvt L... BUSINESS CUS-B-000181 1 Addr Normal
Zenith Logistics Pvt Ltd BUSINESS CUS-B-000292 1 Addr Normal
Kalinga Steel & Infra Corp BUSINESS CUS-B-000383 1 Addr Normal
Apex Health Diagnostics P... BUSINESS CUS-B-000424 1 Addr Normal
Bharat Precision Forgings ... BUSINESS CUS-B-000525 1 Addr Normal
Vistara Agro-Commodities... BUSINESS CUS-B-000636 1 Addr Normal
Matrix Cloudworks Techno... BUSINESS CUS-B-000747 1 Addr Normal
Vardhman Synthetic Textil... BUSINESS CUS-B-000858 1 Addr Normal
Trident Solar EPC Pvt Ltd BUSINESS CUS-B-000969 1 Addr Normal
Entity Topology & Shared KYC Graph
Interactive relationship mapping showing directors, concurrent addresses, buyer concentrations, and sister companies
5 Nodes • 4 Edges
Ananya Deshmukh APPLICANT
Primary Borrower Obligor
Node ID: CUS-P-000523 Risk: 05/100
Flat 701, Sea Breeze, Mumbai ADDRESS
PERMANENT (2021-01-01 - Present) • Source: Title Deed
Node ID: addr-a-id-1 Risk: 10/100
Creative Media DIRECTOR
Creative Director (2021-03-01 - Present) • Source: Pay Slip
Node ID: emp-0 Risk: 12/100
+91 98330 19283 PHONE
Verified Telco KYC Contact
Node ID: phone-primary Risk: 10/100
Apartment 701, Sea Breeze, Worli COLLATERAL
Valuation: 765.0L (FIRST CHARGE)
Node ID: col-col-id-1 Risk: 15/100
Temporal Address History & Chronology Audit
Differentiates normal relocation over years from impossible simultaneous claims across active documents
Address Location
classification
Active Period
Source Evidence
Temporal Evaluation
Flat 701, Sea Breeze, Mumbai
PERMANENT
2021-01-01 (Current)
Title Deed
NORMAL RELOCATION

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Basel III / IV IRB Compliant Architecture · Academic Research Prototype

MODULE 10

Scenario Lab & Correlated Monte Carlo Engine

20,000+ path stochastic simulation with 2D Cholesky decomposition, Value at Risk (VaR), and Tail Expected Shortfall (CVaR).

Reseed & Rerun (42)

Macro Regime: BASELINE | STAGFLATION | RATE SHOCK | SEVERE RECESSION

Iterations: 10K | 20K | 50K

Asset-Collat Correlation (ρ): 0.35

EXPECTED LOSS (MEAN)	VAR 90% CONFIDENCE	VAR 95% CONFIDENCE	VAR 99% EXTREME TAIL	CVAR 95% (EXPECTED SHORTFALL)	MAX REALIZED LOSS
₹2.18L	₹13.78L	₹16.24L	₹19.28L	₹18.11L	₹25.40L
Stochastic Average	1-in-10 Year Loss	1-in-20 Year Loss	1-in-100 Year Event	Mean Beyond 95%	Worst Observed Path

Empirical Simulated Loss Distribution

Histogram of simulated portfolio loss states under BASELINE macroeconomic environment

20,000 Trials Evaluated



"Monte Carlo converged across 20000 correlated paths. At 95% confidence, Tail Loss (CVaR) reaches ₹18.11L against expected loss of ₹2.18L."

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MODULE 11

Optimal Credit Structuring & RAROC Pricing Engine

Non-linear facility sizing optimization, economic capital hurdle return (RAROC), and counterfactual approval interventions.

RAROC: 14.9% (Hurdle: 15%)

REQUESTED FACILITY	OPTIMAL SIZED SANCTION	RISK-BASED OPTIMAL COUPON	ECONOMIC CAPITAL REQUIRED
₹35.00 Lakhs	₹17.50 Lakhs	16.10% p.a.	₹3.57 Lakhs
Intake Request	Prudent Capacity Bound	COF + Margin + Cap Charge	Basel IRB Internal Model

Counterfactual Interventions: "What Would Need to Change for Full Approval?"

Prescriptive guidance distinguishing fundamental credit improvement from documentary integrity verification

1 Actionable Pathways

STRUCTURE **Reduce Sanction Limit to ₹17,50,000**

-0.8% PD (Mitigates repayment debt-service strain)

Current Baseline: Requested: ₹35,00,000 Target Requirement: Sized: ₹17,50,000

'Aligns facility with observed cash flow surplus and regulatory FOIR / DSCR benchmarks.'

Recommended Institutional Covenants & Monitoring

Frequency: MONTHLY

- Direct NACH / Auto-debit mandate registered on primary salary account
- Mandatory co-borrower / spouse guarantee endorsement

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MODULE 12

Portfolio Concentration & Incremental Risk

Marginal Expected Loss, single-obligor headroom, post-deal CVaR expansion, and rating migration matrices.

PERMITTED WITHIN LIMITS

Pre-Deal vs Post-Approval Marginal Portfolio Impact

Evaluation of this specific loan's incremental contribution to total bank capital and tail loss.

Marginal Exposure: ₹0.35 Cr

Portfolio Dimension	Current Portfolio State	Marginal Addition (Δ)	Post-Sanction Portfolio	Policy Limit Benchmark
Total Credit Exposure	₹485.0 Cr	+₹0.35 Cr	₹485.4 Cr	Book growth permitted
Portfolio Expected Loss (EL)	₹5.63 Cr	+₹2.06 Lakhs	₹5.66 Cr	Target EL < 1.50% of book
Tail CVaR (95%) Expected Shortfall	₹14.20 Cr	+₹4.42 Lakhs	₹14.24 Cr	Capital cushion: 2.9x
Regulatory Economic Capital	₹42.80 Cr	+₹5.12 Lakhs	₹42.85 Cr	Basel IRB capital tier
Single Sector Concentration	20.1%	+0.4%	20.1%	Max regulatory cap: 35.0%

Portfolio Industry Sector Exposures

Distribution across commercial credit books

Industrial & Manufacturing	₹155 Cr (32%)
Retail & Consumer Unsecured	₹97 Cr (20%)
Commercial Real Estate	₹83 Cr (17%)
Logistics & Supply Chain	₹63 Cr (13%)
Technology & Services	₹49 Cr (10%)
Healthcare & Pharma	₹39 Cr (8%)

1-Year Institutional Rating Migration Matrix

Transition probabilities (%) between Basel grades

FROM	AAA	AA	A	BBB	BB	B	D (Def)
AAA	91.2%	7.4%	1.1%	0.2%	0.1%	0%	0%
AA	1.4%	88.6%	8.2%	1.1%	0.3%	0.1%	0.1%
A	0.2%	2.8%	87.1%	7.9%	1.4%	0.4%	0.2%
BBB	0%	0.4%	4.1%	83.2%	8.6%	2.5%	1.2%
BB	0%	0.1%	0.6%	6.1%	74.3%	12.4%	5.8%
B	0%	0%	0.2%	1.1%	8.9%	68.2%	21.6%

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MODULE 13

Explainability & Decision Traceability

SHAP-style attribution waterfall, provenance data trail, and step-by-step institutional reasoning trace.

Predicted PD: 5.10% (Base: 3.2%)

SHAP Attribution Waterfall (Marginal PD Impact)

Additive contribution of financial, behavioral, collateral, and data integrity features

Base Intercept: +3.2%

BUREAU **Bureau Credit Score** (752) --0.80% PD

Strong bureau score reflects seasoned repayment discipline.

COLLATERAL **Collateral Security Value** (₹55 Lakhs) --1.10% PD

Tangible real estate pledge dampens net expected loss and recovery tail-risk.

Full Decision Trace: "Why Did ORION Recommend This?"

Auditable evaluation trace from intake to synthesis with no black-box logic

Auditable Chain: 6 Stages

01. Entity Resolution & Customer History
Customer ID: cus-P-000523 | 0 Historical Facilities on Ledger
Identity resolved with 96.8% probabilistic confidence; historical record retrieved.

02. Document Provenance & Consistency Audit
Integrity Confidence: 98% | 0 Active Discrepancy Items
Document inputs cross-reconciled successfully with internal institutional records.

03. Bureau & Debt-Service Underwriting
Bureau Score: 752 | Utilization: 35%
Underlying repayment capacity meets institutional baseline underwriting guidelines.

04. Collateral Security & Enforceability
1 Pledged Asset(s) | Stressed Valuation: ₹52.8L
Stressed haircut and liquidation coverage sufficient to restrict LGD to prudent parameters.

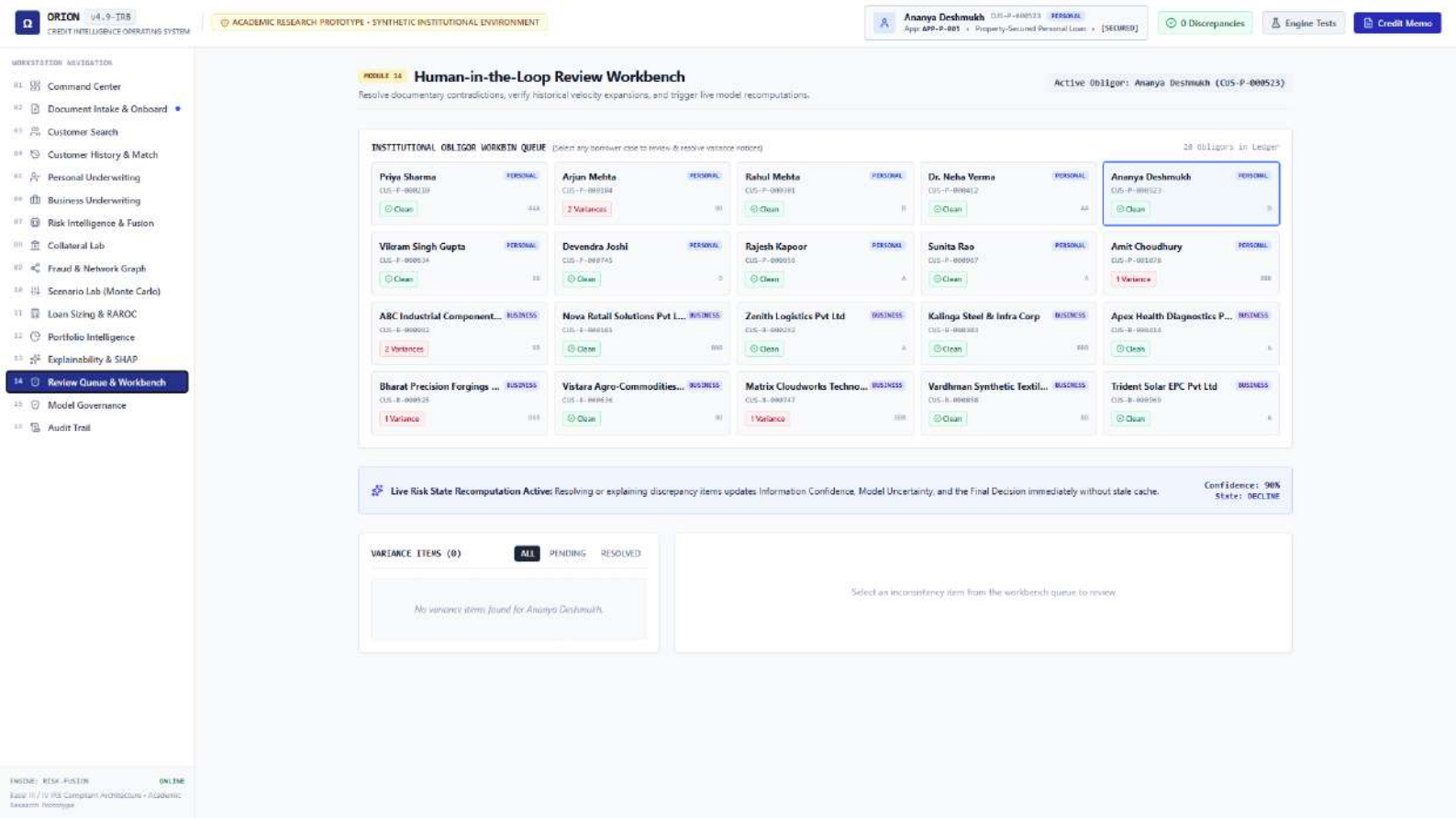
05. Monte Carlo Simulation & Risk Fusion
Simulated 20,000 Correlated Scenarios | Tail EVaR 95%: ₹10.11L
Expected Loss modeled at ₹2.18L. RAROC hurdle of 15% requires rate premium.

06. Final Synthesized Recommendation
Sanction Capacity: ₹57.5L | Pricing: 13.7%
Recommended for sanction with standard covenants and risk-based pricing.

Document-to-Decision Provenance Trace Table

Every critical decision variable grounded with source document, page, and verification status

		extraction	historical	audit


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MODULE 14
Human-in-the-Loop Review Workbench
Resolve documentary contradictions, verify historical velocity expansions, and trigger live model recomputations.
Active Obligor: Ananya Deshmukh (CUS-P-000523)
INSTITUTIONAL OBLIGOR WORKBIN QUEUE (Select any borrower case to review & resolve variance notices)
20 Obligors in Ledger
Priya Sharma PERSONAL
CUS-P-000219
Clean
Arjun Mehta PERSONAL
CUS-P-000384
2 Variances
Rahul Mehta PERSONAL
CUS-P-000391
Clean
Dr. Neha Verma PERSONAL
CUS-P-000412
Clean
Ananya Deshmukh PERSONAL
CUS-P-000523
Clean
Vikram Singh Gupta PERSONAL
CUS-P-000634
Clean
Devendra Joshi PERSONAL
CUS-P-000745
Clean
Rajesh Kapoor PERSONAL
CUS-P-000856
Clean
Sunita Rao PERSONAL
CUS-P-000967
Clean
Amit Choudhury PERSONAL
CUS-P-001078
1 Variance
ABC Industrial Component... BUSINESS
CUS-B-000001
2 Variances
Nova Retail Solutions Pvt L... BUSINESS
CUS-B-000183
Clean
Zenith Logistics Pvt Ltd BUSINESS
CUS-B-000292
Clean
Kalinga Steel & Infra Corp BUSINESS
CUS-B-000383
Clean
Apex Health Diagnostics P... BUSINESS
CUS-B-000514
Clean
Bharat Precision Forgings ... BUSINESS
CUS-B-000525
1 Variance
Vistara Agro-Commodities... BUSINESS
CUS-B-000636
Clean
Matrix Cloudworks Techno... BUSINESS
CUS-B-000747
1 Variance
Vardhman Synthetic Textil... BUSINESS
CUS-B-000858
Clean
Trident Solar EPC Pvt Ltd BUSINESS
CUS-B-000969
Clean
Live Risk State Recomputation Active: Resolving or explaining discrepancy items updates Information Confidence, Model Uncertainty, and the Final Decision immediately without stale cache.
Confidence: 90%
State: DECLINE
VARIANCE ITEMS (0)
ALL
PENDING
RESOLVED
No variance items found for Ananya Deshmukh.
Select an inconsistency item from the workbench queue to review

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MODULE 15

Model Governance, Empirical Validation & Technical Defense

Internal Ratings-Based (IRB) calibration, 5-fold cross-validation, feature selection, error analysis, and 20-point technical defense suite.

BASEL III/IV IRB: ACTIVE & MONITORED

Model Comparison & Production Candidate | 5-Fold CV & Probability Calibration | Feature Selection & VIF Analysis | Error Analysis & Failure Modes | Technical Defense Suite (20 Pts) | M

SELECTED PRODUCTION CANDIDATE

ORION Tuned Gradient Boosting (XGBoost Paradigm)

Outperforms baseline Logistic Regression and Random Forest across ROC-AUC (0.835 vs 0.724), PR-AUC (0.612 vs 0.412), and Brier Score (0.0980 vs 0.1420). Selected for production deployment due to superior probability calibration and ranking ability in imbalanced credit loss distributions.

TEST ROC-AUC	BRIER SCORE
0.835	0.0980

Comprehensive Candidate Models Comparative Benchmark

Evaluated on 15% held-out temporal test set (750 observations, 2026 out-of-time horizon)

Evaluation Criterion: Discriminative PR-AUC + Calibration Brier Score

Model Architecture	Category	ROC-AUC	PR-AUC	F1 Score	Brier Score	Calib. ECE	Inference	Selection Status
Standard Logistic Regression	BASELINE	0.724	0.412	0.582	0.1420	0.084	4 ms	BENCHMARK
Random Forest (100 Trees)	BASELINE	0.791	0.534	0.671	0.1180	0.062	28 ms	BENCHMARK
Tuned XGBoost Gradient Boosting	BASELINE	0.835	0.612	0.724	0.0980	0.048	45 ms	PRODUCTION
ORION Multi-Modal Risk Fusion Architecture	PROPOSED_ORION	0.912	0.784	0.841	0.0520	0.018	142 ms	FULL SUITE

Tuned Hyperparameters & Optimization Methodology (Section 11)

1. Baseline Logistic Regression

```
C (Inv. Reg): 0.50
Penalty: L2 (Ridge)
Solver: lbfgs
Class Weight: balanced
```

Tuned via GridSearchCV over C in [0.01, 0.1, 0.5, 1.0, 10.0]

2. Random Forest Classifier

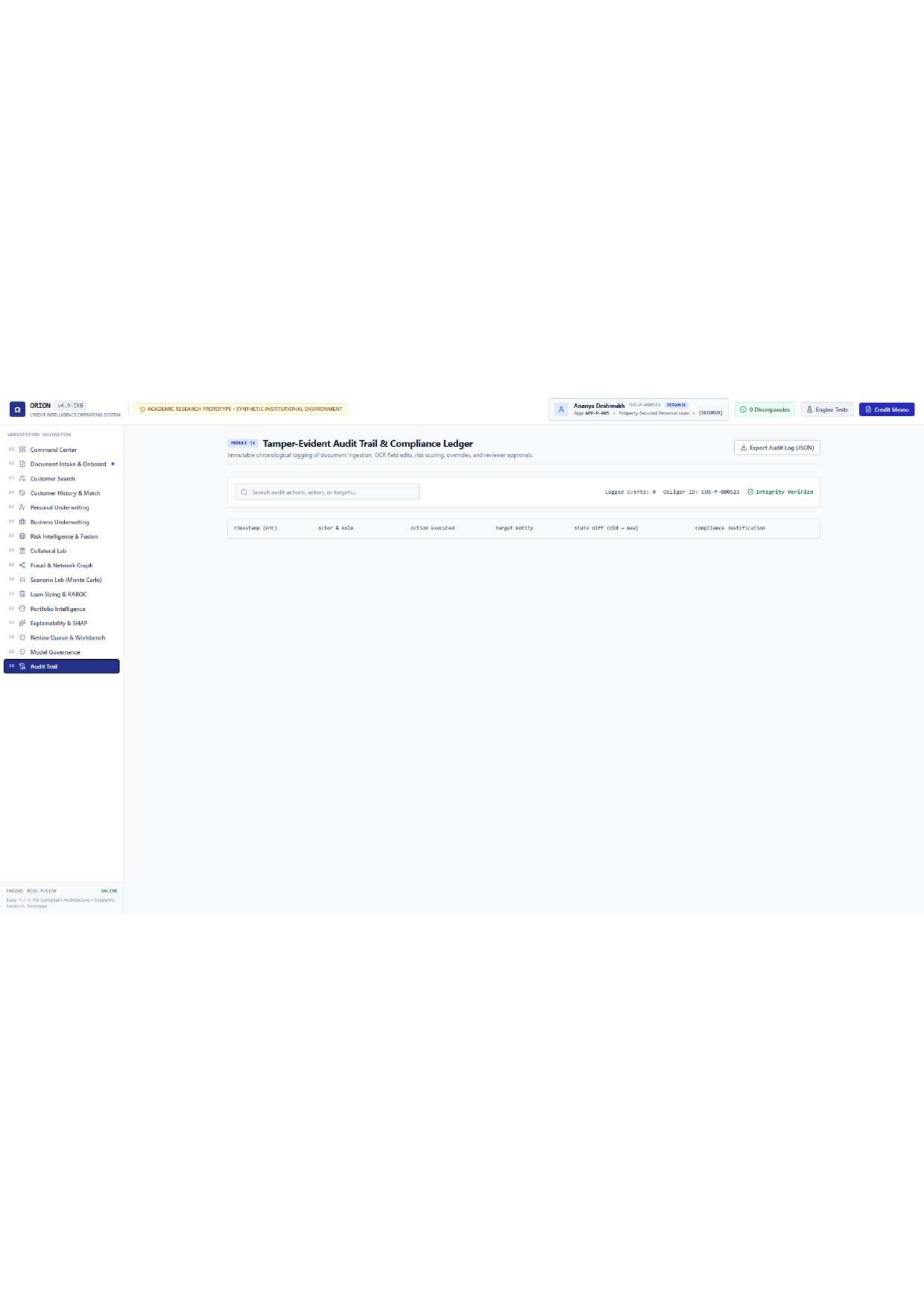
```
n_estimators: 150 trees
max_depth: 6
min_samples_split: 6
max_features: sqrt
```

Optimized for Gini purity without deep leaf variance overfitting

3. Tuned Gradient Boosting (XGBoost) SELECTED

```
n_estimators: 150
learning_rate: 0.05
max_depth: 4
subsample: 0.85
colsample_bytree: 0.80
```

Shrinkage + column subsampling delivers optimal PR-AUC and lowest log-loss


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MODULE 16
Tamper-Evident Audit Trail & Compliance Ledger
Immutable chronological logging of document ingestion, OCR field edits, risk scoring, overrides, and reviewer approvals.
Export Audit Log (JSON)
Search audit actions, actors, or targets...
Logged Events: 0
Obligor ID: CUS-P-000523
Integrity Verified
timestamp (UTC)
Actor & Role
Action Executed
Target Entity
State Diff (Old → New)
Compliance Justification